

Fundamentals of IT Security

IT security doesn't start at large companies – it starts with your everyday behavior. How do you handle passwords? What information do you disclose? Do you open attachments without checking them? Many security issues arise not from technical failure, but because people are too trusting or underestimate risks.



IT Security as Layers of Protection

Passwords

Protect access to your digital spaces

Data Minimization

Protects your privacy and personal information

Caution with Attachments

Protects your device from malware

System Maintenance

Keeps your device up to date and resilient

These layers of protection complement each other – none replaces the other.

CHAPTER 1



Secure Passwords

A password is the key to your digital space. The Federal Office for Information Security (BSI) recommends: Passwords should be **long**, **unique**, and difficult to predict – and should never be reused for multiple services.



Characteristics of Secure Passwords

Long

Length is one of the most important security factors – a long passphrase beats a short, artificially complicated password.

Unique

Use a separate password for every important account. Reusing them enables **credential stuffing**.

Not guessable

No names, birth dates, or simple patterns like "Summer2024".

Not reused

Anyone who knows your email password can often reset all your other accounts.

Password Comparison: Weak vs. Strong

Feature	Insecure	More Secure
Length	6–8 characters	Much longer, as a passphrase
Predictability	Name, birthday, 123456	Random word combination
Reuse	Same password everywhere	Every account has its own
Personal connection	Pet name, favorite team	No guessable references


 Example: **Weak:** Summer2024 → **Stronger:** CoffeeTrainMoonMeadow19 → **Ideal:** a long, random passphrase, for this single service only

Credential Stuffing: Why Password Reuse is Dangerous

If an online shop is hacked and you used the same password there as for your email account, attackers can test this password automatically across many services – this is called **Credential Stuffing**. Your email account is particularly critical: anyone who has access to it can reset passwords for other services.



Passwords in Everyday Life: The Four Principles

1

Long instead of just complicated

A passphrase is often stronger than a short password with special characters

2

Unique instead of reused

Every account deserves its own password

3

Use a password manager

Stores all passwords securely – you only have to remember one strong master password

4

Two-factor authentication

A second form of verification (e.g., app code) protects you even if your password becomes known



CHAPTER 2



Protecting Personal Data

Personal data includes not only your name and address, but also your location, daily routine, photos of your home, travel plans, and answers to security questions. Individually, many pieces of information may seem harmless—together, they create an accurate picture of you that fraudsters and attackers can specifically exploit.

Particularly Sensitive Data Types

Data Type	Example	Why Sensitive
Access credentials	Email, password, PIN	Enable direct account access
Financial data	IBAN, credit card details	Can lead to financial loss
Identity data	ID photo, date of birth	Misused for identity theft
Health data	Diagnoses, medical reports	Very private and prone to misuse
Location data	Live location, travel plans	Reveal habits and absences
Work data	Internal documents	Can endanger you or others

Data minimization in everyday life

The four key questions

- Who really needs to receive this information?
- Why is it being requested?
- Is it necessary to provide everything?
- What could happen if it is shared?

Typical risk situations

Social networks: Never post photos of ID cards, flight tickets, or shipping labels. Background details in photos can reveal addresses or names.

Calls & messages: Scammers often create a sense of urgency: "Act immediately," "Account locked." That's exactly when you should slow down.

Online accounts: Only fill out mandatory fields – if a service doesn't need your date of birth, you don't have to provide it.

How to concretely protect your personal data



Share consciously & sparingly

Only share as much information as is truly necessary. Check privacy settings in apps and on social networks.



Protect sensitive data

Do not share sensitive data via email or chat. Be careful with photos, screenshots, and documents.



Verify trustworthiness

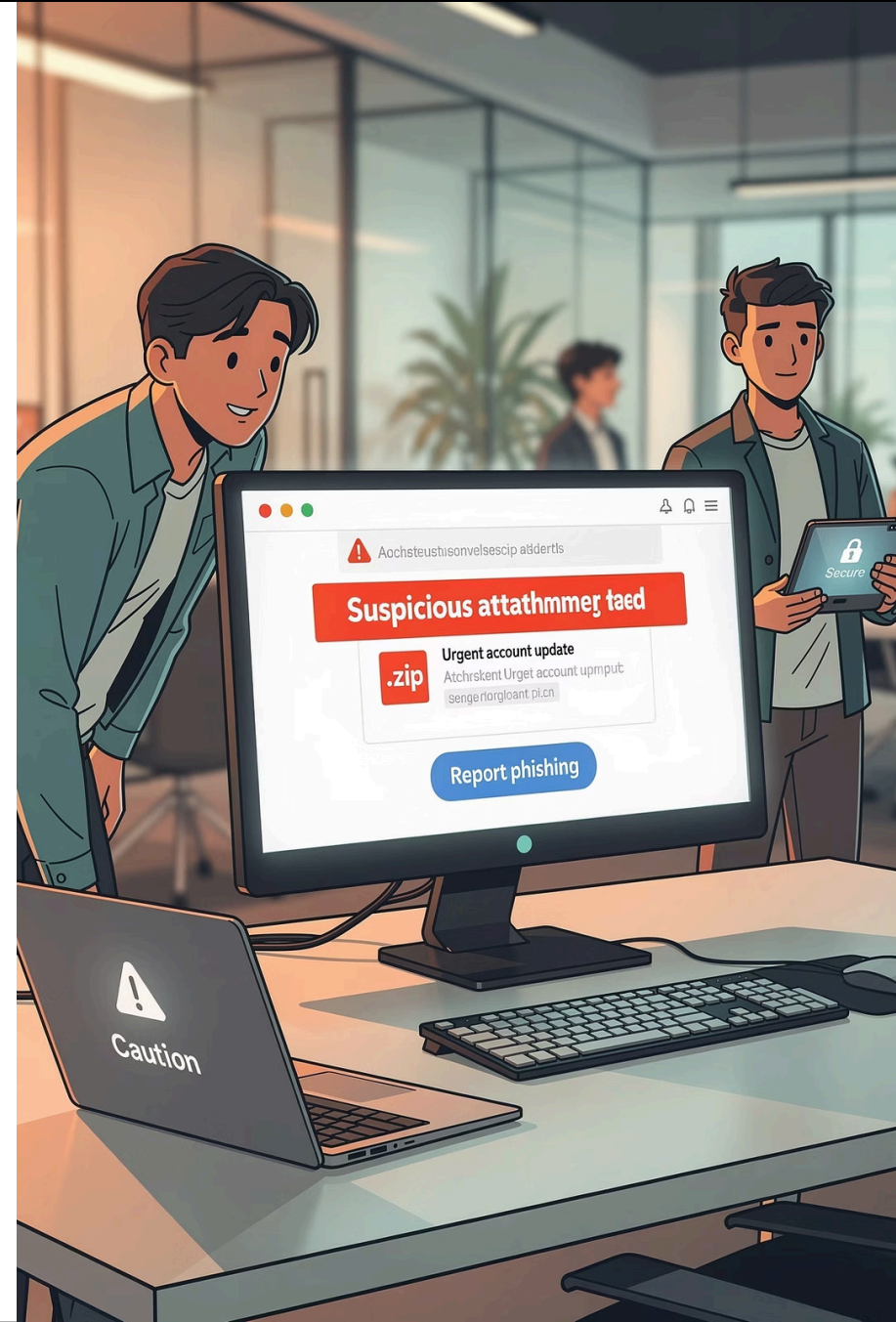
Check whether a website seems trustworthy before entering any data. Use strong passwords and two-factor authentication.

CHAPTER 3



Unknown Attachments – A Classic Attack Vector

Many malware programs do not reach devices through spectacular hacker attacks, but rather quite banally via email. The BSI warns: Phishing messages use pressure, deception, and spoofed senders to trick people into clicking. Sender addresses can be faked – only open an attachment if you have a reason to trust it.



Warning Signs for Suspicious Attachments

→ Unexpected Message

You didn't request anything, but suddenly an invoice, reminder, or job application appears.

→ Suspicious Sender

A slightly altered sender address with swapped letters or an unusual domain.

→ Pressure and Urgency

Words like "urgent," "immediately," or "final notice" are intended to trick you into clicking quickly.

→ Unusual File Type

Files that can execute programs or macros are particularly dangerous. If the extension is hidden, that is already a warning sign.

Check, then open – the 5 questions

1

Do I expect this message?

2

Does the sender really match?

3

Does the content fit the situation?

4

Is the attachment logical in context?

5

Can I verify the sender another way?

☐ If you receive a strange file from a colleague: Do not reply to the suspicious email, but check via another secure channel instead.

SECURITY AWARENESS

What you should never do with attachments

It's precisely that **"just a quick look"** that is dangerous. Many malicious programs require only a single thoughtless click. Many attacks fail simply because people pause briefly instead of clicking reflexively.

⛔ Never

- Open attachments just because the subject line sounds urgent
- Enable macros without knowing exactly why
- Execute unknown file formats
- Dismiss security warnings
- "Take a quick look" even though the message seems strange





CHAPTER 4



System Protection and Maintenance

IT security consists not only of cautious behavior but also of a well-maintained system. Even very attentive users remain vulnerable if protection software is missing or software is outdated. Three pillars must be distinguished: **Antivirus**, **Firewall**, and **Updates**.

Comparing Antivirus, Firewall, and Updates

Security Measure	Main Task	Typical Question
Antivirus	Detect and block malicious software	Is this file suspicious?
Firewall	Control network connections	Is this connection allowed in or out?
Updates	Fix vulnerabilities	Is the software up to date?

None of these measures replaces the others. Effective security is always an **interplay of several layers of protection**: The firewall blocks unwanted connections, the antivirus detects malicious files, and updates prevent known vulnerabilities from being exploited in the first place.



Installing programs safely

Trustworthy sources

- Official manufacturer websites
- Official app stores or package repositories
- Approved internal software sources (company)

Less trustworthy

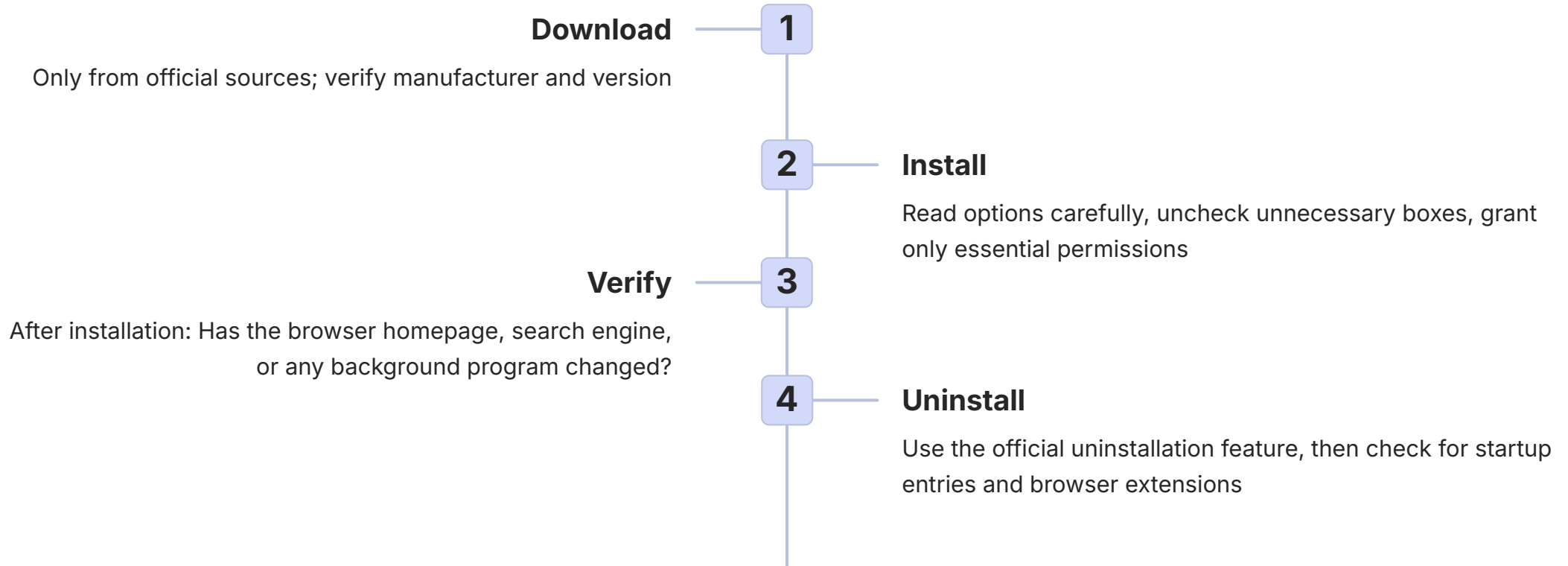
- Dubious download portals
- Email or chat file attachments
- "Cracked" or illegally modified programs

Check before installing

- Do you really need the program?
- Does it come from a trustworthy source?
- Is the program asking for unusually high permissions?
- Is additional software being installed with it?

- ☐ Watch out for checkboxes like "Install recommended additional software" or "Change homepage" – these are often unwanted.

Safe Installation and Uninstallation



The less unnecessary software installed, the smaller your attack surface. Every additional program can introduce bugs or security vulnerabilities.



Why updates are so important

The BSI (Federal Office for Information Security) explicitly points out that updates and patches close security vulnerabilities and **should be installed promptly**. Software can appear to run perfectly normally on the outside while still containing a known security vulnerability internally. Attackers specifically look for such unpatched systems—where the problem is known, but not yet fixed.

Types of Updates and Highly Vulnerable Software

Types of Updates

Type	Meaning
Security update	Closes security vulnerabilities
Bug fix	Resolves bugs or crashes
Feature update	Introduces new features
Driver update	Improves hardware control
Firmware update	Updates routers, printers, SSDs

Highly Vulnerable Software

- Operating systems
- Web browsers
- Email programs
- Office applications & PDF readers
- Apps with internet access
- Routers and network equipment

❏ Many forget the router – it also runs programs with security vulnerabilities.

Handling Updates in Daily Life



Enable Automatically

Enable automatic updates where it makes sense. Install security updates promptly, don't delay them for weeks.



Back Up Before Major Updates

Back up important data, ensure sufficient battery power, and do not start updates in the middle of critical tasks.



Use Official Channels

Avoid dubious "driver updaters" or "system boosters" – the safest path is the manufacturer's official update mechanism.

If a device or program no longer receives security updates, it is not "old but okay" – it is a permanent risk.

The Core Principle: Security is a Process

Passwords

Long, unique, password manager + 2FA

Data Privacy

Data minimization, share consciously, check privacy settings

Attachments

Check, then open – never click reflexively

System Maintenance

Antivirus, firewall, updates – all three together

The most important thing is not just knowing **how** to act, but **why**. Anyone who understands the logic behind IT security makes better decisions automatically in everyday life.